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Subject: Programming with Java	**Semester: III**
ASSIGNMENT No : 17	**Year: AY 2026-27**
TITLE : Swing and Applets	

Problem Statement

Write a Java program to demonstrate the use of Swing and Applets.

Learning Objectives

To develop graphical user interfaces using Swing components and Applets.

Program:

UserRegistration.java

```
// Assignment 17 - Exercise 1: Swing GUI Window Application
import javax.swing.*; // For GUI Components
import java.awt.*; // For Layout
public class UserRegistration
{
   public static void main(String args[]) // Main Method
   {
       JFrame frame = new JFrame("Student Registration Form"); //
Creating top-level window frame
       frame.setSize(400, 300); // Setting frame dimensions
       frame.setLayout(new GridLayout(4, 2, 10, 10)); // Setting grid
layout with spacing
       frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE); // Closing
operation

       // Name
       JLabel lblName = new JLabel("Enter Name: "); // Label
       JTextField txtName = new JTextField(); // Textfield for input

       // Roll No
       JLabel lblRoll = new JLabel("Enter Roll No: "); // Label
       JTextField txtRoll = new JTextField(); // Textfield for input
```

```
        // Branch
        JLabel lblBranch = new JLabel("Enter Branch: "); // Label
        JTextField txtBranch = new JTextField(); // Textfield for input

        // Buttons
        JButton btnSubmit = new JButton("Submit"); // Submit button
        JButton btnReset = new JButton("Clear"); // Clear button

        frame.add(lblName); // Adding name label to frame
        frame.add(txtName); // Adding name textfield to frame
        frame.add(lblRoll); // Adding roll label to frame
        frame.add(txtRoll); // Adding roll textfield to frame
        frame.add(lblBranch); // Adding branch label to frame
        frame.add(txtBranch); // Adding branch textfield to frame
        frame.add(btnSubmit); // Adding submit button to frame
        frame.add(btnReset); // Adding reset button to frame

        frame.setVisible(true); // Displaying frame window
    }
}
```

AppletDemo.java

```
// Assignment 17 - Exercise 2: Basic Java Applet
// HTML Run Code: <applet code="AppletDemo.class" width="350"
height="200"></applet>
// Executing via Swing as Applet support has been deprecated.
import java.applet.Applet; // Applet package
import java.awt.*; // AWT graphics and fonts
import javax.swing.JFrame; // Swing JFrame container

public class AppletDemo extends Applet // Applet base class
{
    public void init() // Applet initialization lifecycle method
    {
        setBackground(Color.LIGHT_GRAY); // Setting background color
    }
    public void paint(Graphics g) // Paint lifecycle method for rendering
graphics
    {
        g.setColor(Color.BLUE); // Drawing color
        g.setFont(new Font("Arial", Font.BOLD, 16)); // Text font
        g.drawString("Welcome to Java Applet Programming!", 30, 60);
```

```
        g.setColor(Color.RED); // Drawing color
        g.drawRect(30, 80, 280, 50); // Rectangle outline
        g.drawString("Symbiosis Institute of Technology", 45, 110);
    }

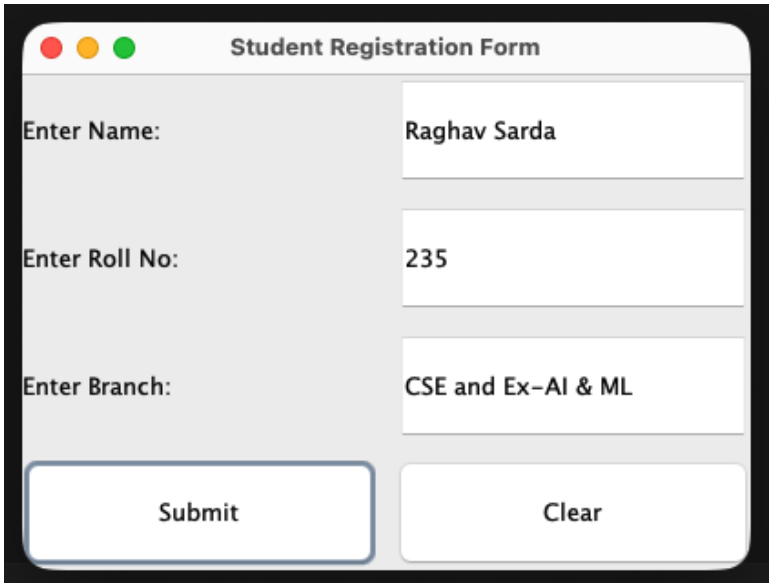
    public static void main(String args[]) // Main Method
    {
        JFrame frame = new JFrame("Applet Viewer"); // Creating top-level
window frame
        AppletDemo applet = new AppletDemo(); // Creating instance of
AppletDemo

        applet.init(); // Calling applet lifecycle initialization
        applet.start(); // Starting applet execution

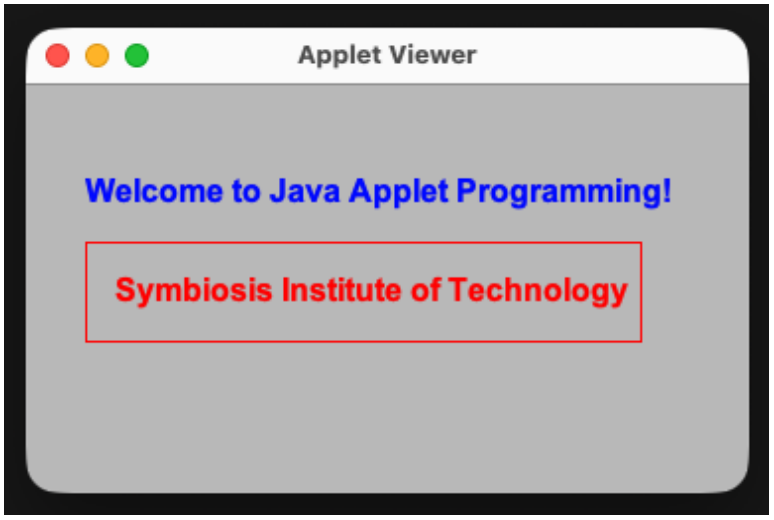
        frame.add(applet); // Adding applet to frame container
        frame.setSize(365, 235); // Setting frame dimensions matching
applet bounds
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE); // Closing
operation
        frame.setLocationRelativeTo(null); // Centering frame on screen
        frame.setVisible(true); // Displaying frame window
    }
}
```

Output:

UserRegistration.java



AppletDemo.java



Conclusion:

In this assignment, the concepts of graphical user interface design using Java Swing components and Applets were successfully implemented. Swing provided lightweight, platform-independent GUI elements like frames, labels, and text fields for interactive window applications, while the Applet demonstration (using Swing package, as Applet has been deprecated) illustrated direct graphical rendering within embedded environments.